Department of Computer Science & Engineering
OPERATING SYSTEMS
UE24CS242B

ASSIGNMENT-4

Github link->
https://github.com/sahil10187/PES2UG24CS281-pes-vcs

Q NO.	Name:MOHAMMED SAHIL S SRN:PES2UG24CS281 SECTION:E

1 A	`sahil10187@LAPTOP-VFO07PD8:/mnt/d/SAHIL/PES/2nd_year/sem2/OS/assignment4/PES2UG24CS281-pes-vcs$ ./test_objects`
	`Stored blob with hash: d58213f5dbe0629b5c2fa28e5c7d4213ea09227ed0221bbe9db5e5c4b9aafc12`
	`Object stored at: .pes/objects/d5/8213f5dbe0629b5c2fa28e5c7d4213ea09227ed0221bbe9db5e5c4b9aafc12`
	`PASS: blob storage`
	`PASS: deduplication`
	`PASS: integrity check`
	
	`All Phase 1 tests passed.`

1 B	`sahil10187@LAPTOP-VFO07PD8:/mnt/d/SAHIL/PES/2nd_year/sem2/OS/assignment4/PES2UG24CS281-pes-vcs$ find .pes/objects -type f`
	`.pes/objects/25/ef1fa07ea68a52f800dc80756ee6b7ae34b337afedb9b46a1af8e11ec4f476`
	`.pes/objects/2a/594d39232787fba8eb7287418aec99c8fc2ecdaf5aaf2e650eda471e566fcf`
	`.pes/objects/d5/8213f5dbe0629b5c2fa28e5c7d4213ea09227ed0221bbe9db5e5c4b9aafc12`

2 A	`sahil10187@LAPTOP-VFO07PD8:/mnt/d/SAHIL/PES/2nd_year/sem2/OS/assignment4/PES2UG24CS281-pes-vcs$ ./test_tree`
	`Serialized tree: 139 bytes`
	`PASS: tree serialize/parse roundtrip`
	`PASS: tree deterministic serialization`
	
	`All Phase 2 tests passed.`

2 B	`sahil10187@LAPTOP-VFO07PD8:/mnt/d/SAHIL/PES/2nd_year/sem2/OS/assignment4/PES2UG24CS281-pes-vcs$ find .pes/objects -type f`
	`.pes/objects/25/ef1fa07ea68a52f800dc80756ee6b7ae34b337afedb9b46a1af8e11ec4f476`
	`.pes/objects/2a/594d39232787fba8eb7287418aec99c8fc2ecdaf5aaf2e650eda471e566fcf`
	`.pes/objects/d5/8213f5dbe0629b5c2fa28e5c7d4213ea09227ed0221bbe9db5e5c4b9aafc12`

3 A

```
sahil10187@LAPTOP-VFO07PD8:/mnt/d/SAHIL/PES/2nd_year/sem2/OS/as
signment4/PES2UG24CS281-pes-vcs$ ./pes init
Initialized empty PES repository in .pes/
```

```
sahil10187@LAPTOP-VFO07PD8:/mnt/d/SAHIL/PES/2nd_year/sem2/OS/as
signment4/PES2UG24CS281-pes-vcs$ ./pes add test_objects
sahil10187@LAPTOP-VFO07PD8:/mnt/d/SAHIL/PES/2nd_year/sem2/OS/as
signment4/PES2UG24CS281-pes-vcs$ ./pes add test_tree
sahil10187@LAPTOP-VFO07PD8:/mnt/d/SAHIL/PES/2nd_year/sem2/OS/as
signment4/PES2UG24CS281-pes-vcs$ ./pes add tree.c
```

```
sahil10187@LAPTOP-VFO07PD8:/mnt/d/SAHIL/PES/2nd_year/sem2/OS/assignment4/PES2UG24CS281-pes-vc
s$ ./pes status
Staged changes:
  staged:     Makefile
  staged:     README.md
  staged:     commit.c
  staged:     commit.h
  staged:     commit.o
  staged:     index.c
  staged:     index.h
  staged:     index.o
  staged:     object.c
  staged:     object.o
  staged:     pes
  staged:     pes.c
  staged:     pes.h
  staged:     pes.o
  staged:     test_objects
  staged:     test_objects.c
  staged:     test_objects.o
  staged:     test_sequence.sh
  staged:     test_tree
  staged:     test_tree.c
  staged:     test_tree.o
  staged:     tree.c
  staged:     tree.h
  staged:     tree.o
```

3 B

```
sahil10187@LAPTOP-VFO07PD8:/mnt/d/SAHIL/PES/2nd_year/sem2/OS/assignment4/PES2UG24CS281-pes-vc
s$ cat .pes/index
100755 52c63e82b1130c3f797394121ddcfb4a5433b8dbf3a1642a7b7520fc1cbda7a8 1776750469 1446 Makef
ile
100755 1e3f8140b075d9dfba3ae870b21ce566ccb2b29438aa746742278886c402285f 1776751859 36113 READ
ME.md
100755 3e10f1092e0f01cf015552d34fe1d48071a8d958041c91ba8013716a014e81b5 1776749982 7501 commi
t.c
100755 9d0cc0889f086edabd586c368f23879a1eceb1d4b03b83d4c4e221d086afdccf 1776746938 2346 commi
t.h
100755 e71a9f37ee2ad00367c6561d53ee4a7df2f6ffc9cbb17a8cb38b3748c5b769f0 1776751883 8440 commi
t.o
100755 0bdf90c22d016e097718a716f342bd8f26a90c440572709aa05909bd1f580ed6 1776750196 9764 index
.c
100755 865015a838a5554d99dc9572e2c44058d1b4eb1f1040f75e80690ba53ad20fdb 1776746938 1799 index
.h
100755 aee645d94d0a090258ad6448779ce67ced58d178acd282fe7c4a3a8d9733e2ae 1776751883 9008 index
.o
100755 653b3d36cf7b2d53b39478459392f423a620a6cabec8105da2db4ab0b49cda05 1776750873 5520 objec
t.c
100755 ccaaa2c729331807599d50ca2c38fe8c5ebfcd3511c09608eda4166e51c70638 1776751883 7992 objec
t.o
100755 a394452c4c1006b84f8554a3e8ae8c12dda432ab0ef0594bfd10aa586e79b221 1776751884 40200 pes
100755 06fbfccffb34fc618332b92362781ac60b660715a3128808a13370dbfb39a9c3 1776746938 4073 pes.c
100755 9912a5af91606587dca377790a431619ea194013729d9533da0b83333920a1e9 1776746938 2471 pes.h
100755 e95829cbebfc4ff105b206ece482623e221adfa1973ba6a858531b6d74739708 1776751883 8920 pes.o
100755 dd6c1275eb49eac021a01b19036a0e9ad392da7039c1f7add67de9f42e7c1ad1 1776751884 22056 test
_objects
100755 9b1c67aa520087a038704a30771fe1c9e2a396b95e7abcdba3476fb169ac6ec6 1776746938 2634 test_
```

4 A

<table>
<tr><td></td><td>

```
sahil10187@LAPTOP-VFO07PD8:/mnt/d/SAHIL/PES/2nd_year/sem2/OS/assignment4/PES2UG24CS281-pes-vc
s$ ./pes commit -m "first commit"
Committed: 81471aa91a90... first commit
sahil10187@LAPTOP-VFO07PD8:/mnt/d/SAHIL/PES/2nd_year/sem2/OS/assignment4/PES2UG24CS281-pes-vc
s$ ./pes commit -m "second commit"
Committed: 84c37bee58b0... second commit
sahil10187@LAPTOP-VFO07PD8:/mnt/d/SAHIL/PES/2nd_year/sem2/OS/assignment4/PES2UG24CS281-pes-vc
s$ ./pes commit -m "third commit"
Committed: 83171f9b77a1... third commit
sahil10187@LAPTOP-VFO07PD8:/mnt/d/SAHIL/PES/2nd_year/sem2/OS/assignment4/PES2UG24CS281-pes-vc
s$ ./pes log
commit 83171f9b77a10a71ec7cb5cc6eff7fafd4c2cecaaf4ca12af90aa55a9a5a070e
Author: PES User <pes@localhost>
Date:   1776753129

    third commit

commit 84c37bee58b09e242953e713db7e41b02d529183b70e674b8309d3eab8acfdfc
Author: PES User <pes@localhost>
Date:   1776753121

    second commit

commit 81471aa91a900510d41acab923fdcca23e29c5ca849bc7f04b397400f1f521f0
Author: PES User <pes@localhost>
Date:   1776753112

    first commit
```

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<tr><td>4 B</td><td>

```
sahil10187@LAPTOP-VFO07PD8:/mnt/d/SAHIL/PES/2nd_year/sem2/OS/assignment4/PES2UG24CS281-pes-vc
s$ find .pes -type f | sort
.pes/HEAD
.pes/index
.pes/objects/06/fbfccffb34fc618332b92362781ac60b660715a3128808a13370dbfb39a9c3
.pes/objects/0b/df90c22d016e097718a716f342bd8f26a90c4405727709aa05909bd1f580ed6
.pes/objects/14/5ed1209ee4b31af7aeba91321d400414231daaa55a7bfcd8a644f76e1de48c
.pes/objects/1e/3f8140b075d9dfba3ae870b21ce566ccb2b29438aa746742278886c402285f
.pes/objects/25/ef1fa07ea68a52f800dc80756ee6b7ae34b337afedb9b46a1af8e11ec4f476
.pes/objects/2a/594d39232787fba8eb7287418aec99c8fc2ecdaf5aaf2e650eda471e566fcf
.pes/objects/30/1784349754fedc835be3ca066dbfd5c16cb1ffbf6539e86be514ba37fe6bf7
.pes/objects/3e/10f1092e0f01cf015552d34fe1d48071a8d958041c91ba8013716a014e81b5
.pes/objects/52/c63e82b1130c3f797394121ddcfb4a5433b8dbf3a1642a7b7520fc1cbda7a8
.pes/objects/5a/041b06606e38ce263434e0e86f18196c0c3eb6b967cf01ee9e25a9cbaf0b43
.pes/objects/65/3b3d36cf7b2d53b39478459392f423a620a6cabec8105da2db4ab0b49cda05
.pes/objects/65/df4fd62f3b96e93f33b3b3924c21ebe5331bf54770d8fa7c9a5018efaf3008
.pes/objects/81/471aa91a900510d41acab923fdcca23e29c5ca849bc7f04b397400f1f521f0
.pes/objects/83/171f9b77a10a71ec7cb5cc6eff7fafd4c2cecaaf4ca12af90aa55a9a5a070e
.pes/objects/84/c37bee58b09e242953e713db7e41b02d529183b70e674b8309d3eab8acfdfc
.pes/objects/86/5015a838a5554d99dc9572e2c44058d1b4eb1f1040f75e80690ba53ad20fdb
.pes/objects/8d/47f4b7eae4a79480357fba470202732287179abd258823611b6e343eba6992
.pes/objects/8d/c02f292fa889bdc5ba09c6a394771bce1eed6c7b1b8cb922167d867ce92844
.pes/objects/99/12a5af91606587dca377790a431619ea194013729d9533da0b83333920a1e9
.pes/objects/9b/1c67aa520087a038704a30771fe1c9e2a396b95e7abcdba3476fb169ac6ec6
.pes/objects/9d/0cc0889f086edabd586c368f23879a1eceb1d4b03b83d4c4e221d086afdccf
.pes/objects/a3/94452c4c1006b84f8554a3e8ae8c12dda432ab0ef0594bfd10aa586e79b221
.pes/objects/ae/e645d94d0a090258ad6448779ce67ced58d178acd282fe7c4a3a8d9733e2ae
.pes/objects/cc/aaa2c729331807599d50ca2c38fe8c5ebfcd3511c09608eda4166e51c70638
```

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<tr><td>4 C</td><td>

```
sahil10187@LAPTOP-VFO07PD8:/mnt/d/SAHIL/PES/2nd_year/sem2/OS/assignment4/PES2UG24CS281-pes-vc
s$ cat .pes/refs/heads/main
83171f9b77a10a71ec7cb5cc6eff7fafd4c2cecaaf4ca12af90aa55a9a5a070e
sahil10187@LAPTOP-VFO07PD8:/mnt/d/SAHIL/PES/2nd_year/sem2/OS/assignment4/PES2UG24CS281-pes-vc
s$ cat .pes/HEAD
ref: refs/heads/main
```

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<tr><td>FINAL</td><td></td></tr>
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```
sahil10187@LAPTOP-VFO07PD8:/mnt/d/SAHIL/PES/2nd_year/sem2/OS/assignment4/PES2UG24CS281-pes-vcs$ make test-integration
=== Running integration tests ===
bash test_sequence.sh
=== PES-VCS Integration Test ===

--- Repository Initialization ---
Initialized empty PES repository in .pes/
PASS: .pes/objects exists
PASS: .pes/refs/heads exists
PASS: .pes/HEAD exists

--- Staging Files ---
Status after add:
Staged changes:
  staged:     file.txt
  staged:     hello.txt

Unstaged changes:
  (nothing to show)

Untracked files:
  (nothing to show)


--- First Commit ---
Committed: 5f5b17b60560... Initial commit

Log after first commit:
commit 5f5b17b6056015c176e873fbc58e19259d66dcd175411bc52fa5b504de476b45
Author: PES User <pes@localhost>
Date:   1776753294

    Initial commit


--- Second Commit ---
Committed: cbf926a00f4b... Update file.txt

--- Third Commit ---
Committed: 584eac325d17... Add farewell

--- Full History ---
commit 584eac325d17115ea8263dc82dc301499ac7bf6a3d2e534be060c8b2b00849a5
Author: PES User <pes@localhost>
Date:   1776753294

    Add farewell

commit cbf926a00f4bae86766b601bcd8ed4b76b237e806ddc8041ba4e799938211f9b
Author: PES User <pes@localhost>
Date:   1776753294

    Update file.txt

commit 5f5b17b6056015c176e873fbc58e19259d66dcd175411bc52fa5b504de476b45
Author: PES User <pes@localhost>
Date:   1776753294

    Initial commit


--- Reference Chain ---
HEAD:
ref: refs/heads/main
refs/heads/main:
584eac325d17115ea8263dc82dc301499ac7bf6a3d2e534be060c8b2b00849a5

--- Object Store ---
Objects created:
10
.pes/objects/0b/d69098bd9b9cc5934a610ab65da429b525361147faa7b5b922919e9a23143d
.pes/objects/10/1f28a12274233d119fd3c8d7c7216054ddb5605f3bae21c6fb6ee3c4c7cbfa
.pes/objects/58/4eac325d17115ea8263dc82dc301499ac7bf6a3d2e534be060c8b2b00849a5
.pes/objects/58/a67ed1c161a4e89a110968310fe31e39920ef68d4c7c7e0d66957975337f50d
.pes/objects/5f/5b17b6056015c176e873fbc58e19259d66dcd175411bc52fa5b504de476b45
.pes/objects/ab/5824a9ec1ef505b5480b3e37cd50d9c80be55012cabe0ca572db4959788299
.pes/objects/b1/7b838c5951aa88c09635c5895ef7e08f7fa1974d901ce282f30e08de0ccd92
.pes/objects/cb/f926a00f4bae86766b601bcd8ed4b76b237e806ddc8041ba4e799938211f9b
.pes/objects/d0/7733c25d2d137b7574be8c5542b562bf48bafeaa3829f61f75b8d10d5350f9
.pes/objects/db/07a1451ca9544dbf66d769b505377d765efae7adc6b97b75cc9d2b3b3da6ff
```

5.1	**pes checkout <branch> works by updating .pes/HEAD to point to the new branch and reading its commit hash from .pes/refs/heads/, then replacing the working directory files to match that commit's tree; this is complex because it must safely handle modified, added, or deleted files without overwriting user changes.**
5.2	**A dirty working directory is detected by comparing the current file contents with hashes stored in the index, and if a file is modified and also differs from the target branch's version, checkout must refuse to prevent data loss.**
5.3	**In detached HEAD state, commits are created but not linked to any branch, so they can become unreachable; however, they can be recovered by using the commit hash or by creating a new branch pointing to that commit.**
6.1	**Garbage collection works by starting from all branch heads, traversing all reachable commits, trees, and blobs, marking them**

	using a hash set, and deleting any unmarked (unreachable) objects; in large repositories, this may involve visiting hundreds of thousands of objects.
6.2	**Running garbage collection during a commit is dangerous because it can delete objects that are created but not yet referenced, leading to inconsistency; Git avoids this using locking and safe write mechanisms to ensure objects are fully linked before being exposed.**